

HEART FAILURE: ROUTINE CARE

The patient with heart failure has leg swelling and difficulty breathing which worsens on lying down/with effort. **A doctor must confirm the diagnosis and refer the patient for specialist assessment.**

Give urgent attention to the patient with heart failure and any of:

- Chest pain → 37.
- Rapid worsening of symptoms
- Respiratory rate ≥ 30 or difficulty breathing
- BP $< 90/60$
- New wheeze

Manage and refer urgently:

- Sit patient up and if oxygen saturation $< 94\%$, give 40% face mask oxygen (6-8L/min).
- If systolic BP > 90 : give **furosemide** 40mg slow IV. If no response after 30 minutes, give another 80mg IV. If good response, give 40mg IV after 2-4 hours.
- If systolic BP > 90 : give sublingual **isosorbide dinitrate** 5mg even if there is no chest pain. Repeat once if pain relief needed. Repeat after 4 hours.
- If BP $\geq 180/130$: give single dose **enalapril** 10mg orally.

Assess the patient with heart failure

Assess	When to assess	Note
Symptoms	Every visit	Manage symptoms as on symptom pages. If fainting/blackouts, refer same day.
Family planning	Every visit	Assess patient's contraceptive needs → 154. If pregnant or planning pregnancy, refer for specialist care.
Alcohol/drug use	Every visit	In the past year, has patient: 1) drunk ≥ 4 drinks ¹ /session, 2) used illegal drugs or 3) misused prescription or over-the-counter medications? If yes to any → 142.
Weight	Every visit	Assess changes in fluid balance by comparing with weight when patient least symptomatic.
BP and pulse	Every visit	If known hypertension → 133. If not, check BP: if $\geq 140/90$ → 132. If new irregular pulse, refer same day.
Palliative care	At diagnosis, if deteriorating	If disabling shortness of breath at rest on maximum treatment or ≥ 5 admissions in the past 6 months, also give palliative care → 170.
Creatinine (eGFR) and potassium	At diagnosis, 6 monthly	• If starting/increasing dose of enalapril/spironolactone: also check at 2 weeks (if eGFR < 60) or 4 weeks (if eGFR ≥ 60). • If creatinine increases by $> 20\%$, eGFR < 30 or potassium > 5.0, stop enalapril/spironolactone and discuss with specialist.
Other blood tests	At diagnosis	Check Hb, TSH and if not known diabetes, check glucose → 17. If abnormal, discuss with specialist. Test for HIV → 110.

Advise the patient with heart failure

- Advise to adhere to treatment even if asymptomatic. Advise regular exercise within limits of symptoms. Help the patient to manage his/her CVD risk → 129.
- Advise to restrict salt to $< \frac{1}{2}$ a teaspoon/day and fluids to 1.5L/day (6 cups). If possible, advise to monitor weight daily. If s/he gains ≥ 2 kg in 2 days, advise to return to clinic.

Treat the patient with heart failure

- Give **influenza vaccine** 0.5mL IM yearly. Check that patient is up to date with his/her COVID-19 vaccine.
- Aim to have patient on steps 1 and 2. Add step 3 if patient has ongoing symptoms on steps 1 and 2. If uncontrolled on steps 1-3, refer to specialist for digoxin.

Step	Medication	Dose	Note
1	Give hydrochlorothiazide or furosemide and enalapril	25-50mg daily	Use if mild heart failure and eGFR ≥ 60 . Avoid in liver disease. Use with caution in gout, previous skin cancer.
		Start 40mg daily. If needed, increase every 2-3 days until symptoms improve, up to 250mg/day.	• Use if significant heart failure symptoms or eGFR < 60. Once improved, consider switch to hydrochlorothiazide if eGFR ≥ 60. • If > 80mg needed, give half dose 12 hourly.
		Start 2.5mg 12 hourly. If needed, increase up to 10mg 12 hourly.	Avoid if pregnant, previous angioedema, aortic stenosis, hypertrophic obstructive cardiomyopathy, renal artery stenosis.
2	Add carvedilol	Start 3.125mg 12 hourly. If tolerated, double dose every 2 weeks until symptoms improve, up to 25mg 12 hourly.	• Start once on optimal dose of enalapril. Avoid atenolol in heart failure. • Avoid if severe fluid overload, BP $< 90/60$, asthma. Avoid or decrease dose if pulse < 60.
3	Add spironolactone	25mg daily	Monitor potassium and kidney function. Avoid if eGFR < 30 or potassium > 5 . Stop potassium supplements.

¹One drink is 1 tot of spirits, or 1 small glass (125mL) of wine or 1 can/bottle (330mL) of beer.